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SUPERSEDING
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FEDERAL SPECIFICATION

# CALKING GUN, BULK AND CARTRIDGE COMPOUNDS

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers calking guns having a steel barrel. They are of the hand-operated type generally used for sealing door and window frames with gun-type calking compound to make the building weatherproof (see 6.1).

1.1.1 *Federal specification coverage.*—Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specification, or which are commercially available, but are intended to cover the types, etc., which are generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and styles.*—Calking guns covered by this specification shall be of the following types, styles, and minimum capacities, as specified (see table I and 6.2):

*Note.*—The capacities listed below are for the gun. For cartridge capacity, see 6.8.

Type I. — For bulk and cartridge compounds with interchangeable washers.

Style 1.—With either smooth or ratchet plunger rod.

¾ pint bulk capacity, 2" x 7" cartridge.
1 pint bulk capacity 2" x 8½" cartridge.

Style 2.—With ratchet plunger rod.

¾ pint bulk capacity, 2" x 7" cartridge.
1 pint bulk capacity, 2" x 8½" cartridge.
1 1/16 pint bulk capacity, 2" x 10" cartridge.
1 5/16 pint bulk capacity, 2" x 10" cartridge.
1½ pint bulk capacity, 2" x 10" cartridge.

Type II.—For bulk compounds.

Style 1.—With either smooth or ratchet plunger rod.

¾ pint bulk capacity (21 cubic inches minimum).
1 pint bulk capacity (28 cubic inches minimum).

Style 2.—With ratchet plunger rod.

¾ pint bulk capacity (21 cubic inches minimum).
1 pint bulk capacity (28 cubic inches minimum).
1 1/16 pint bulk capacity (30 cubic inches minimum).
1 5/16 pint bulk capacity (37 cubic inches minimum).
1½ pint bulk capacity (42 cubic inches minimum).
1 quart bulk capacity (57 cubic inches minimum).

## 2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:



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*Federal Specifications:*

TT–C–598—Compound, Calking, Plastic (For Masonry and Other Structures).
PPP–B–566 — Boxes, Folding, Paperboard.
PPP–B–585—Boxes, Wood, Wirebound.
PPP–B–591—Boxes, Fiberboard, Wood-Cleated.
PPP–B–601—Boxes, Wood, Cleated-Plywood.
PPP–B–621—Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636—Boxes, Fiber.
PPP–B–676 — Boxes, Set-Up, Paperboard.

*Federal Standards:*

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

*Military Specifications:*

MIL–B–4229 — Boxes; Paperboard, Metal-Stayed.
MIL–B–10377—Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL–L–10547 — Liners, Case, Waterproof.

*Military Standards:*

MIL–STD–129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Illustration.** — The illustration shown herein is for the convenience of identification and is not intended to preclude the purchase of calking guns which are otherwise in accordance with the requirements of this specification.

**3.2 Design.**—The calking guns covered by this specification shall be of the portable, hand-operated type. When the operator actuates the trigger the plunger rod shall apply pressure to the plunger, and the plunger in turn shall force the calking compound through the discharge nozzle. The plunger rod shall provide for quick adjustment of the plunger anywhere within the barrel. The gun shall operate efficiently, and without leakage, with grade 1 plastic calking compound as specified in Federal Specification TT–C–598. The guns shall conform to the minimum requirements of table I and be similar to figure 1.

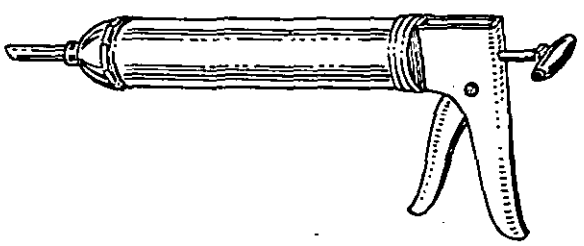


FIGURE 1.—Calking gun.

**3.3 Marking.**—Each gun shall be marked in a plain and permanent manner with the name of the manufacturer or with a trademark of such known character that the name of the manufacturer may be readily determined.



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TABLE I

| Type | Style | Bulk capacity (minimum) | Cartridge sizes [1] | | Capacity of gun (minimum) | Weight of gun (minimum) | |
|---|---|---|---|---|---|---|---|
| | | | Diameter | Length | | | |
| | | *Pints* | *Inches* | *Inches* | *Cu. In.* | *Pounds* | *Ounces* |
| I | 1 | ¾ | .2 | 7 | 21 | 2 | 0 |
| I | 1 | 1 | 2 | 8½ | 28 | 2 | 4 |
| I | 2 | ¾ | 2 | 7 | 21 | 2 | 0 |
| I | 2 | 1 | 2 | 8½ | 28 | 2 | 4 |
| I | 2 | 1⅛ | 2 | 10 | 30 | 2 | 6 |
| I | 2 | 1⅜ | 2 | 10 | 37 | 2 | 7 |
| I | 2 | 1½ | 2 | 10 | 42 | 2 | 8 |
| II | 1 | ¾ | -- | ---- | 21 | 2 | 0 |
| II | 1 | 1 | -- | ---- | 28 | 2 | 4 |
| II | 2 | ¾ | -- | ---- | 21 | 2 | 0 |
| II | 2 | 1 | -- | ---- | 28 | 2 | 4 |
| II | 2 | 1⅛ | -- | ---- | 30 | 2 | 6 |
| II | 2 | 1⅜ | -- | ---- | 37 | 2 | 7 |
| II | 2 | 1½ | -- | ---- | 42 | 2 | 8 |
| II | 2 | 2 | -- | ---- | 57 | 3 | 8 |

[1] For cartridge capacities see 6.3.

3.4 Interchangeability. — All replaceable parts shall be constructed to definite standards, tolerances, and clearances in order that any such part may be replaced without requiring modification.

3.5 Finish.

3.5.1 *Nonmilitary agencies.*—Unless otherwise specified (see 6.1), all steel and iron parts (except springs) shall have a durable coating for preventing corrosion.

3.5.2 *Military agencies.* — For military agencies all steel and iron parts (except springs) shall have a cadmium or any electroplated finish.

3.6 Construction.—The construction shall be free from any characteristics or defects that might render the gun unsuitable or inefficient for the purpose intended. The gun shall consist essentially of a barrel, front cap, rear cap and handle, plunger rod, small components, and discharge nozzle.

3.6.1 *Barrel.*—The barrel shall be cylindrical in shape and of steel at least 0.050 inch thick.

3.6.2 *Front cap.*—The front cap shall be of steel at least 0.050 inch thick, either conical or semiconical in shape, and provide a means for interchanging discharge nozzles. Unless otherwise specified (see 6.2), the front cap shall be furnished with either machine cut threads or other means of interchangeability such as a bayonet joint.

3.6.3 *Rear cap and gun handle.*—The rear cap and gun handle shall be of steel at least 0.050 inch thick. The rear cap and gun handle may be either separate components securely fastened together or an integral component. Unless otherwise specified (see 6.2), the rear cap shall be either fastened to the barrel or provided with threads or other easy means for assembly and disassembly of the two components. The handle shall be of such size as to provide a firm grip for one-hand operation of the gun.



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3.6.4 *Plunger rod.*—The plunger rod shall operate efficiently with each stroke of the trigger and shall be furnished with a handle to provide quick adjustment of the plunger anywhere within the barrel. The rod shall be of sufficient length to discharge the contents of the gun barrel.

3.6.5 *Trigger.*—The trigger shall be of steel and of such length and actuating distance as to provide for one-hand operation of the gun.

3.6.6 *Plunger.*—The plunger for bulk type compounds shall be leather cup-type or equally suitable design and construction with a means of expanding the cup to a neat slip fit in the gun barrel to form a leakproof joint. Means shall be provided for changing the plunger components. The plunger for cartridge type compounds shall operate efficiently in 2-inch nominal diameter cartridge (see 6.3).

3.6.7 *Small components.* — All small applicable components such as gaskets, pawls, dogs, fasteners, springs, etc. shall be of a good commercial quality. Applicable parts shall be heat-treated when necessary to prevent excessive wear.

3.6.8 *Discharge nozzle.*—The calking guns shall be furnished with at least one detachable discharge nozzle of the style and size normally furnished by the contractor. When specified (see 6.2), additional detachable discharge nozzles shall be furnished from the following styles as specified:

Style 1—Round, 1/8, 1/4, 3/8, or 1/2 inch discharge.

Style 2—Triangular, 1/4, 3/8, or 1/2 inch discharge.

Style 3—Flat, 3/8, 1/2, 3/4, or 1 inch discharge.

Style 4 — Oblong, 1/8 or 3/16 inch discharge.

Style 5—Seaming, 1 1/4 inches discharge threaded for end of barrel.

Style 6—Round or flat, 3/4 inch discharge threaded for end of barrel.

3.7 **Type I.**—Type I guns shall be provided with interchangeable plunger components for the use of both cartridge and bulk type compounds. The plunger components for the use of cartridges, shall be of such design as to function satisfactorily in interchangeable models of various manufacturer's cartridges that fit into a 2 inch nominal and minimum inside diameter gun barrel (see 6.3). The barrel for type I guns shall be of 2 inch nominal (and minimum) inside diameter. The length of the barrel shall be of sufficient length to hold at least the minimum amount of bulk compound and the minimum length of cartridge specified in table I.

3.7.1 *Style 1.*—Type I, style 1 calking guns shall be provided with either a smooth plunger rod or a plunger rod having ratchet teeth.

3.7.2 *Style 2.*—Type I, style 2 calking guns shall be provided with a plunger rod having ratchet teeth.

3.8 **Type II.**—Type II guns shall be suitable for bulk type compounds. At the option of the contractor, type I guns may be bid in competition with the corresponding styles of type II guns. Type II guns shall not be governed by any cartridge size and therefore shall be acceptable if they are well-proportioned as to diameter and length, and if they meet the applicable capacity requirements of table I and other applicable requirements of this specification. The barrel of type II guns shall have a minimum capacity for bulk compound as specified in table I.

3.8.1 *Style 1.*—Type II, style 1 calking guns shall be provided with either a smooth plunger rod or a plunger rod with ratchet teeth.

3.8.2 *Style 2.*—Type II, style 2 calking guns shall be provided with a plunger rod having ratchet teeth.



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3.9 Workmanship.—Workmanship shall be of a good quality. Burrs and sharp edges shall be removed from such parts that might cause injury to the operator's hands during use or disassembly of the gun. The gun shall operate efficiently without leakage, without undue tightness of movable parts, or without skipping of the ratcheting mechanism.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

4.1.1 *Inspection lot.*—All guns of the same type, style and size offered for delivery at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.1.2 *Sampling for visual and dimensional inspection.*—A random sample of guns shall be selected from each inspection lot of material offered for Government inspection in accordance with Military Standard MIL–STD–105 at Inspection Level III for lot sizes of 40 and under, Inspection Level II for lot sizes 41 to 300 and Inspection Level I for lots over 301 as shown in table II. Acceptable quality level shall be 1.5 percent defective.

TABLE II.—*Sampling for visual and dimensional inspection AQL (approx.) = 1.5 percent defective*

| Number of guns in lot | Number of guns in sample | Acceptance number (defective) | Rejection number (defective) |
|---|---|---|---|
| 110 and under | 10 | 0 | 1 |
| 111 to 800 | 35 | 1 | 2 |
| 801 to 1300 | 50 | 2 | 3 |
| 1301 to 3200 | 75 | 3 | 4 |
| 3201 and over | 150 | 5 | 6 |

4.1.3 *Sampling for lot acceptance.*—A random sample of guns shall be selected from each inspection lot offered for Government testing in accordance with the appendix to Military Standard MIL–STD–105 at Inspection Level L–4 as shown in table III. The acceptable quality level shall be 4.0 percent defective.

TABLE III.—*Sampling for tests AQL (approx.) = 4.0 percent defective*

| Number of guns in lot | Number of guns in sample | Acceptance number (defective) | Rejection number (defective) |
|---|---|---|---|
| 8 and under | 1 | 0 | 1 |
| 9 to 180 | 2 | 0 | 1 |
| 181 to 500 | 3 | 0 | 1 |
| 501 to 800 | 5 | 0 | 1 |
| 801 to 1800 | 7 | 0 | 1 |
| 1801 and over | 10 | 1 | 2 |

4.2 Visual and dimensional inspection.—Each of the sample guns selected in accordance with table II shall be visually and dimensionally inspected to verify compliance with the requirements of this specification. Any gun in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective guns in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3 Lot acceptance tests.—Each of the sample guns selected in accordance with table III shall be tested in accordance with 4.4 to verify compliance with the requirements of this specification. Any sample gun which does not meet the requirements for any of these characteristics shall be rejected, and if the number of nonconforming guns in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.4 Test procedures.

4.4.1 *Capacity test.*—Each of the sample guns shall be tested for capacity compliance. The type I gun shall be tested with a commercial cartridge of the applicable size specified in table I. The type I and type II guns



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shall be tested for the minimum cubic contents in accordance with table I.

4.4.2 *Performance.*—The sample guns shall be inspected for performance by calking around one or more window or door frames. The calking compound shall be forced into small crevices and also lay a uniform fillet of compound in the intersecting angle of the window frame and its adjacent surface. The gun shall then be inspected to insure that there is no leakage of compounds at the various joints. The plunger cup washer for bulk compounds shall be removed to insure that there is no leakage into the gun barrel behind the plunger. The plunger rod feed mechanism shall not skip when the trigger of the gun is actuated. The guns shall also be tested for quick adjustment of the plunger.

4.5 **Rejected lots.**—Rejected lots may be offered again for Government inspection provided the contractor has repaired or removed all nonconforming guns. The Government Inspector shall again select and examine samples from such resubmitted lots to verify compliance with the specification requirements.

## 5. PREPARATION FOR DELIVERY

(For nonmilitary agencies, Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Packaging.

5.1.1 *Level A.*—Unless otherwise specified (see 6.2), guns shall be individually packaged in folding paperboard, set-up, metal-stayed paperboard, corrugated or solid fiberboard boxes conforming to Federal Specification PPP–B–566, PPP–B–676, PPP–B–636 or Military Specification MIL–B–4229.

5.1.2 *Level C.*—The guns shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.* — The guns, packaged as specified in 5.1.1 shall be packed in snug-fitting wood-cleated solid fiberboard, nailed wood, wood-cleated veneer, wood wirebound, wood-cleated plywood or fiberboard boxes conforming to Federal Specification PPP–B–591, PPP–B–621, PPP–B–585 (class 2 or 3), PPP–B–601 (overseas type), PPP–B–585 or Military Specification MIL–B–10377. Shipping containers shall be provided with waterproof case liners conforming to class 2 of Military Specification MIL–L–10547 and sealed in accordance with the appendix thereto. Gross weight of wood and wood-cleated boxes shall not exceed 200 pounds; of fiberboard boxes, 70 pounds.

5.2.2 *Level B.* — The guns packaged as specified in 5.1, shall be packed in snug-fitting wood-cleated fiberboard, nailed wood, wood wire-bound, wood-cleated plywood, wood-cleated veneer, corrugated or solid fiberboard boxes conforming to the requirements of Federal Specification PPP–B–591, PPP–B–621, PPP–B–585 (class 1), PPP–B–601 (domestic type), Military Specification MIL–B–10377 (domestic type), or Federal Specification PPP–B–636. Gross weight of wood and wood-cleated boxes shall not exceed 200 pounds. Gross weight of fiberboard boxes shall not exceed the permissible weight limitation of the applicable box specification.

5.2.3 *Level C.* — The guns packaged as specified in 5.1.2, shall be packed in containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Nonmilitary requirements.*—In addition to any special marking specified in the contract or order (see 6.2), each unit and



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intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military requirements.*—In addition to any special marking required by the contract or order, shipments shall be marked in accordance with Military Standard MIL–STD–129.

## 6. NOTES

6.1 Intended use.—Type I guns are recommended for occasional calking jobs because of the cleanliness in handling of the cartridges and also the ease provided in loading the gun. Type II guns are recommended for large calking jobs because of the economy in purchasing bulk compound.

*Note.*—Type I corresponding styles of guns may be bid in competition with the type II guns because both types are adapted to bulk compounds.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein and procurement documents should specify the type and capacity of gun required.

(*a*) Type, style and capacity (see 1.2 and table I).
(*b*) Front cap means of interchangeability, if different, so state (see 3.6.2).
(*c*) If rear cap has to be detachable, so state (see 3.6.3).
(*d*) Style, size, and quantity of additional nozzles, if required (see 3.6.8).
(*e*) Levels of preservation, packaging and packing (see section 5).
(*f*) Marking, if different (see 5.3).
(*g*) When cartridges are to be procured, state with or without spout (see 6.3).

6.3 Cartridges.—Note that in tables I and IV the cubic contents of the cartridges and the cubic contents of bulk compound for the same size gun are not equivalent, the reason being that some of the bulk compound is displaced in the the gun barrel by the cartridge shell and its end components. The shell of the cartridge shall be a rolled cardboard cylinder approximately 0.050 inch thick. The cartridge shall be provided with a metal inverted cup washer that serves as a plunger component in forcing the compound through the cartridge cylinder when the ratcheting mechanism of the gun is actuated. The opposite end of the cartridge shall be provided with a metal end cap having an opening for the discharge of the compound. The opening shall be provided with a spout or without a spout, as specified (see 6.2). The cartridges for the guns covered by this specification shall be in accordance with table IV.

TABLE IV.—*Cartridge sizes*

| Outside diameter + 0 — 1/16 | Length + 1/16 — 1/4 | Cartridge capacity | |
|---|---|---|---|
| *Inches* | *Inches* | *Rated* | *Approximate cubic inch equivalent* |
| 2 | 7 | 1/8 pint | 18 |
| 2 | 8 1/8 | 1/10 gallon | 23 |
| 2 | 10 | 1 pint | 29 |

6.4 Illustration.—The illustration shown herein is for the convenience of identification and is not intended to preclude the purchase of calking guns which are otherwise in accordance with the requirements of this specification.

6.5 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.



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6.6 Transportation description.—Transportation description applicable to this item is:

Calking guns, hand.
Carload minimum weight 30,000 pounds.
Mechanics' hand tools, not otherwise indexed.
Truckload minimum weight 30,000 pounds.

**Notice.** — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
Army—E Q T
Navy—Sh MC Y
Air Force.

Copies of this specification may be purchased for 5 cents each.



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